

DEXTER FERNANDES

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SUMMARY

Computer Vision Engineer with 3+ years building and deploying production-grade deep learning systems, from CNN architecture adaptation and training through to real-time edge and cloud inference. Owned the full ML lifecycle for 24/7 perception in physically demanding field environments, including custom CUDA/TensorRT inference acceleration and evaluation of modern Vision Transformer and foundation-model architectures. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar – AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation in desert field conditions.
- Adapted YOLOv5 CNN detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna-driven hyperparameter optimisation and Weights & Biases tracking across runs.
- Designed adversarial-condition training experiments covering occlusion, lighting variation and motion artefacts, benchmarking robustness on F1 and failure rate to reach a 15% F1 gain in 3 months.
- Ran TIDE error analysis on COCO-format production data to split live detector failures by error type, directing dataset fixes and threshold tuning at the dominant one.
- Benchmarked YOLOv5 against EfficientDet on a repeatable harness, quantifying accuracy, latency and stability pre and post optimisation to catch regressions before release.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Presented performance findings to leadership, project managers and rig operators, turning production signals into rollout and optimisation decisions.

LivNSense Technologies – Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage CNN detector for construction safety in Texas, owning production ML pipelines end to end from ingestion through deployment and field validation, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Added ByteTrack multi-object tracking over detections to hold worker and vehicle identities across frames on a moving construction vehicle, cutting duplicate incident alerts.
- Evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for annotation review, foreground separation and failure analysis on industrial camera feeds.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Set experimentation standards, reviewed model performance and mentored junior engineers, guiding optimisation strategy for production deployments.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Developed and evaluated an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Implemented a custom C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine for custom-layer deployment.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size, evaluating CNN and Vision Transformer approaches for edge deployment suitability.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess Vision Transformer and foundation-model viability for detection, segmentation and depth estimation under edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic, fusing LiDAR and RGB sensing for multi-modal SLAM and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations to support system selection.

EDUCATION

University of Bristol – MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time multi-modal perception system in C++17 and ROS2, fusing Velodyne LiDAR, RealSense depth and radiological sensing on a mobile robot for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Ran the full perception stack on an NVIDIA Jetson AGX Xavier under JetPack, working within the platform's compute and memory limits for on-device operation.

Visvesvaraya National Institute of Technology – BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
CV & Deep Learning	Convolutional neural networks (CNNs), Vision Transformers (RT-DETR, EfficientViT, SegFormer), object detection, image classification, semantic and instance segmentation, object tracking, optical flow, anomaly detection, defect detection, data augmentation, YOLO (v5/v8/v10/v26), EfficientDet, Depth Anything v2, PyTorch, TensorFlow, Keras, OpenCV, torchvision, HuggingFace
Model Training	Training strategy design, hyperparameter optimisation (Optuna), performance evaluation, ablation studies, hypothesis testing, Bayesian optimisation, Weights & Biases, Jupyter
Edge & Acceleration	TensorRT (custom CUDA plugins, dynamic batching), OpenVINO, ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, layer fusion, Intel VTune, NVIDIA Jetson AGX Xavier
Real-Time Video	NVIDIA DeepStream 6.x/9.0, GStreamer, FFmpeg, RTSP, multi-stream video analytics, multi-camera synchronisation, ByteTrack, NvDCF, NvSORT
Production & Cloud	Docker, containerised microservices, CI/CD, Git, pytest, unittest, gtest, Microsoft Azure, REST API design, structured logging, observability, production monitoring, Linux
MLOps	MLflow, DVC, Apache Airflow, CVAT, automated labelling, synthetic data generation, benchmarking harnesses, drift tracking, TIDE, COCO annotation format
Robotics & Multi-Modal	ROS2, multi-modal sensor fusion (LiDAR, RGB-D), point cloud processing (PCL, Open3D), Ceres, g2o, GTSAM
Professional	Technical leadership, mentoring, Agile and Scrum, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage – Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)